Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/12/2023
PI Name:	Komatsu, David Ph.D.		
Project Title:	Effects of Simulated Microgravity on Osteoblast-Osteoclast Coupling and Therapeutic Countermeasures		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Stony Brook	**State:**	NY
Zip Code:	11794-8181	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	02/11/2020	**End Date:**	12/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per NSSC information. (Ed. 12/15/22) NOTE: End date changed to 07/26/2023 per NSSC information. (Ed. 12/15/22) NOTE: End date changed to 07/26/2022 per NSSC information. (Ed. 11/19/21)		

Key Personnel Changes/Previous PI:	NOTE (Dec 2022): Per PI -- Dr. Sardar Uddin's involvement with the project as a formal consultant has ended, though he still helps with discussions (Ed. 12/15/22). NOTE (Dec 2021): Dr. Sardar Uddin remains a Consultant and CoInvestigator on the current award, with Dr. David Komatsu serving as Principal Investigator (PI). The studies are being completed by a pharmacology graduate student, Isaiah Taylor, who joined the laboratory in the spring of 2021. These studies will serve as the core of his Ph.D. dissertation. NOTE (March 2020): New principal investigator is Dr. David Komatsu, who replaces previous PI - Dr. Sardar Uddin; Dr. Uddin remains on the project as a CoInvestigator. Kaitlin Farrell completed her work as a research technician in July of 2023. However, she has been hired with different funds and is still providing some assistance to Isaiah Taylor with the outcomes analyses, as well as overall with lab management.
COI Name (Institution):	Uddin, Sardar Ph.D. (State University of New York, Stony Brook (Stony Brook University))
Grant/Contract No.:	80NSSC18K1479
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: Continuation of "Effects of Simulated Microgravity on Osteoblast-Osteoclast Coupling and Therapeutic Countermeasures--PI Uddin" with new Principal Investigator (PI) Dr. David Komatsu, who replaced previous PI, Dr. Sardar Uddin, in February 2020. Objective: Astronauts lose 1-1.5% of their bone mass per month during low Earth orbit missions. Spaceflight profiles of 2 years or longer, as projected for Mars missions, will significantly increase bone fragility, potentially compromising the ability of astronauts to perform their mission tasks in transit and on Mars. To facilitate human exploration of Mars, it is therefore essential to understand the underlying causes of microgravity-induced bone loss and develop effective countermeasures to sustain bone structure and mechanical integrity under microgravity conditions. Our long-term goal is to protect the skeletal health of astronauts in space environments so they can perform their mission operations. The objective of the current proposal is to determine the effects of microgravity on osteoblast-osteoclast coupling and communication in order to identity a countermeasure for microgravity-induced decoupling. Our central hypothesis is that simulated microgravity will impair osteoblast-osteoclast coupling and that a bisphosphonate (alendronate) and/or anti-RANKL antibody (denosumab) treatment will mitigate microgravity-induced osteoblast-osteoclast decoupling. The rationale for this proposal is the fact that microgravity promotes osteoclastic bone resorption and impairs osteoblastic bone formation. It is expected that physiological osteoblast-osteoclast coupling is impaired and that denosumab, a receptor activator of nuclear factor-?B ligand (RANKL) inhibitor, will restore osteoblast-osteoclast coupling and mitigate microgravity-induced bone loss to a greater degree than alendronate, an anti-catabolic agent. Specific Aims and Methods: The central hypothesis will be tested by completing the following two specific aims. 1. Determine the effects of simulated microgravity on osteoblast-osteoclast coupling and communication utilizing in vitro co-cultures. Our working hypothesis is that microgravity will alter osteoblast-osteoclast communication processes. Pathway analyses will be conducted by assessing the differential expression of mRNA (e.g., RNA sequencing, or "RNA-seq"; quantitative polymerase chain reaction, or "qPCR") and protein (i.e., enzyme-linked immunosorbent assay, or "ELISA" tests; Western blots), as well as histology and histomorphometry. Combined Confocal Raman Scanning Electron Microscopy (CRSEM) will be used to determine functional impairment of bone resorption and formation. 2. Assess the efficacy of denosumab relative to alendronate as a countermeasure to microgravity-induced osteoblast-osteoclast decoupling in vitro. Our working hypothesis is that denosumab will be more efficacious in restoring microgravity-induced impairment of osteoblast-osteoclast coupling as it functions by blocking excess RANKL, while alendronate only affects resorption. Osteoblast-osteoclast communication will be assessed using proteomic and genetic analysis and function restoration will be assessed using histology, histomorphometry, and CRSEM. Significance and Outcomes: Bone remodeling is a complex process that is tightly regulated by osteoblast-osteoclast coupling and communication. Microgravity impairs this process by increasing osteoclast-associated resorption and decreasing osteoblastic bone formation. In order to continue to expand our crewed spaceflight program, it is essential to determine the effects of microgravity on osteoblast-osteoclast coupling and develop targeted countermeasures to sustain bone health in microgravity. At the completion of the study, we expect to characterize the effects of microgravity on osteoblast-osteoclast coupling, and identify communication pathways associated with microgravity, as well as establish the efficacy of alendronate and/or denosumab as potential therapeutics to restore osteoblast-osteoclast communication and function. The outcomes of this study will fill NASA's recognized knowledge gap and guiding question in microgravity effects on osteoblast-osteoclast communication and evaluate potential therapeutics to the aberrant microgravity-induced bone loss in astronauts (AH2, 3, CMB1, 2, 6). [Ed. note Dec 2021: NASA research gaps have changed since original proposal. For information on Animal Biology (formerly AH), or Cell and Molecular Biology (CMB), see Space Biology program: https://]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research is highly innovative as it uses, for the first time, creative osteoblast-osteoblast communication on the bone chip to study bone formation and resorption. The study will use innovative experimental approaches to study bone formation and bone resorption, resulting in the first comprehensive in vitro hylomorphic analysis of bone formation and resorption relative to changing gravity and osteoblast-osteoclast activity. The successful completion of research will allow us to develop new approaches to studies of osteoblast-osteoclast communication in different gravitational and pathological conditions. Studying the changes in osteoblast-osteoclast coupling will enable us to identify new therapeutic targets to facilitate treatment of bone loss in space and Earth based conditions.

Task Progress:	In the current review period, we have completed our biomechanical testing runs for the primary human osteoblasts and the primary human osteoclasts. For both cell types, the cells were seeded on cylindrical mixed cortical/cancellous bone blocks. For each run, 12 bone blocks were subjected to simulated microgravity (SMG) by rotating the cultures at ~30 RPM, with an additional 12 blocks cultured in the same vessels with no rotation. After 28 days, the samples were testing in unconstrained compression. SMG impaired the biomechanical properties of blocks seeded with both osteoblasts and osteoclasts, with significant reductions seen in stiffness, energy to failure, yield force, ultimate force, and failure force. We also completed two time course studies of blocks seeded with MC3T3 cells, a mouse osteoblast line. These were used to visualize cell growth on the blocks under SMG and static conditions weekly for 28 days via scanning electron microscopy (SEM) and confocal imaging after live/dead staining. These two imaging modalities both showed increase in cell growth throughout the blocks and only small numbers of dead cells. Now that these protocols have been established, they will be repeated using the primary osteoblasts and osteoclasts. A final study was performed to quantify cell growth. In this study, MC3T3 cells were seeded and grown in static culture and analyzed weekly up to 28 days. Total DNA was then extracted and quantified using spectroscopy. These data showed an initial drop in DNA content at the first week, followed by a leveling off and slight increase on days 14 and 21, and finally a large increase on day 28. These studies will now be repeated with primary osteoblasts and osteoclasts under SMG.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Abstracts for Journals and Proceedings	Taylor IT, Uddin SMZ, Komatsu DE. "Osteoblast and osteoclast activity alters bone biomechanical properties under simulated microgravity." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Poster# GR28. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Taylor I, Uddin SMZ, Komatsu DE. "Exploring the impact of microgravity on osteoblast and osteoclast function." 12th Annual Musculoskeletal Repair and Regeneration Symposium, New York, NY, October 25, 2023. Abstracts. 12th Annual Musculoskeletal Repair and Regeneration Symposium, New York, NY, October 25, 2023. , Oct-2023
Abstracts for Journals and Proceedings	Taylor I, Uddin SMZ, Komatsu DE. "Detrimental effects of microgravity on osteoblast and osteoclast activity." ASBMR 2023 (American Society for Bone and Mineral Research) Annual Meeting, Vancouver, BC, October 13-16, 2023. Abstracts. ASBMR 2023 Annual Meeting, Vancouver, BC, October 13-16, 2023. Poster. Abstract ID #SAT-186. , Oct-2023